Accounting for additive genetic mutations on litter size in Ripollesa sheep[1]

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ABSTRACT: Little is known about mutational variability in livestock, among which only a few mutations with relatively large effects have been reported. In this manuscript, mutational variability was analyzed in 1,765 litter size records from 404 Ripollesa ewes to characterize the magnitude of this genetic source of variation and check the suitability of including mutational effects in genetic evaluations of this breed. Threshold animal models accounting for additive genetic mutations were preferred to models without mutational contributions, with an average difference in the deviance information criterion of more than 5 units. Moreover, the statistical relevance of the additive genetic mutation term was checked through a Bayes factor approach, which showed that the models with mutational variability were 8.5 to 22.7 times more probable than the others. The mutational heritability (percentage of the phenotypic variance accounted for by mutational variance) was 0.6 or 0.9%, depending on whether genetic dominance effects were accounted for by the analytical model. The inclusion of mutational effects in the genetic model for evaluating litter size in Ripollesa ewes called for some minor modifications in the genetic merit order of the individuals evaluated, which suggested that the continuous uploading of new additive mutations could be taken into account to optimize the selection scheme. This study is the first attempt to estimate mutational variances in a livestock species and thereby contribute to better characterization of the genetic background of productive traits of interest.

Key words: animal breeding, genetic variability, litter size, mutation, Ripollesa breed

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INTRODUCTION

As discussed by Goddard (2001), the standard infinitesimal model used for genetic evaluations of a given phenotypic trait of interest simplifies its genetic background to a large number of genes with small and additive effects. These additive effects are also related to the genetic variability present in the founder population (Henderson, 1973), in which mutations that arise in further generations are assumed not to exist. Sources of variation such as nonadditive effects or mutation are commonly omitted, although multiple analytical approaches accounting for these biological phenomena have been developed in recent decades (Wray, 1990; Varona and Misztal, 1999).

Mutation is the ultimate source of polygenic variation because it spontaneously contributes new alleles to the genetic pool (Hill, 1982b). The continuous uploading of new mutations has been revealed as a relevant source of new additive genetic variability in both mice (Keightley and Hill, 1992; Casellas and Medrano, 2008) and other animals (Caballero et al., 1991), which has motivated experiments to monitor responses to long-term artificial selection (Keightley, 1998; Hill, 2007). Nevertheless, little is known about the contribution of new mutations in livestock, which are probably limited to a few loci with major effects on each phenotype (MacLennan and Phillips, 1992; Kambadur et al., 1997; Davis et al., 2004). Although estimates of mutational heritability in experimental species have ranged from 10^{-2} to 5×10^{-4} (Lynch, 1988; Houle et al., 1996),

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this variable has never previously been estimated in livestock.

Given that litter size is one of the most important traits in meat sheep production (Olesen et al., 1994), many studies have focused on genetic background, yielding a wide range of moderate- to low-heritability estimates (Waldron and Thomas, 1992; Fogarty, 1995). Within this context, the purpose of the present study was to evaluate the impact of new mutations on additive genetic variability associated with litter size in Ripollesa sheep.

MATERIALS AND METHODS

All animal protocols were approved by the Universitat Autònoma of Barcelona Animal Care and Use Committee. Moreover, analyses were performed on existing data obtained under standard farm management procedures without any additional requirements. The records analyzed were systematically registered by the staff of the experimental farm of the Universitat Autònoma of Barcelona (Bellaterra, Spain) between 1986 and 2007.

Field Data Source

Litter size data were collected for the Ripollesa experimental flock kept at the Universitat Autònoma of Barcelona. The Ripollesa is a medium-sized sheep breed (rams, 75 to 90 kg of BW; ewes, 50 to 65 kg of BW) with an approximate census of 30,000 ewes in northeastern Spain (Casellas et al., 2007c). This breed is reared under semiintensive Mediterranean conditions for the production of Pascual-type lambs (22 to 24 kg of BW at slaughter; Casellas et al., 2007c). Litter size (1.2 to 1.7 lambs per litter; Casellas et al., 2007b) and lamb mortality before slaughter (approximately 10%; Casellas et al., 2007d) tend to be moderate.

Analyses were performed on 1,765 litter size records (1.49 ± 0.01 lambs per litter) from 404 Ripollesa ewes with single (53.8%) and multiple (twins, 44.1%; triplets, 1.9%; quadruplets, 0.2%) deliveries. These data were collected from the experimental Ripollesa flock at the University Autònoma de Barcelona. Since its foundation in 1986 from the acquisition of purebred individuals from 3 Ripollesa farms (Torre Marimon farm, Caldes de Montbui, Spain; Torrebonica farm, Terrassa, Spain; SEMEGA farm, Monells, Spain), this flock has been closed, with the exception of 2 rams purchased in 1997 (Casellas et al., 2007a) from the Torre Marimon farm. Ewes lambed once per year between September and November (ewe-lambs lambed between December and February; Casellas et al., 2007b) and all relevant data were recorded at lambing (ram, ewe, date of lambing, and number of lambs born). The full pedigree included 22 rams and 541 ewes, 98 of which lacked known ancestors (17.4%). The number of daughters with phenotypic data per ram ranged between 1 and 26, with an average of 10.3 ± 1.5 daughters per ram.

Genetic Analyses and Model Comparison

Analytical Models. Litter size data were analyzed through threshold animal models (Wright, 1934) implemented by Bayesian inference. The vector of liabilities (**u**, see below), a set of values sampled from an unobservable continuous Gaussian process (Sorensen et al., 1995; Van Tassell et al., 1998), was assumed to be influenced by systematic (**b**), permanent environmental (**p**), and genetic effects (**G**) as follows:

$$\mathbf{u} = \mathbf{Xb} + \mathbf{Z}_1\mathbf{p} + \mathbf{G} + \mathbf{e},$$

where $\mathbf{X}$ and $\mathbf{Z}_1$ were appropriate incidence matrices, and $\mathbf{e}$ was the vector of the residuals. Eight alternative models were defined by assuming $\mathbf{G} = \mathbf{0}$ (model 0), $\mathbf{G} = \mathbf{Z}_2\mathbf{a}$ (model A), $\mathbf{G} = \mathbf{Z}_2\mathbf{d}$ (model D), $\mathbf{G} = \mathbf{Z}_2\mathbf{m}$ (model M), $\mathbf{G} = \mathbf{Z}_2\mathbf{a} + \mathbf{Z}_2\mathbf{d}$ (model AD), $\mathbf{G} = \mathbf{Z}_2\mathbf{a} + \mathbf{Z}_2\mathbf{m}$ (model AM), $\mathbf{G} = \mathbf{Z}_2\mathbf{d} + \mathbf{Z}_2\mathbf{m}$ (model DM), and $\mathbf{G} = \mathbf{Z}_2\mathbf{a} + \mathbf{Z}_2\mathbf{d} + \mathbf{Z}_2\mathbf{m}$ (model ADM). Note that $\mathbf{0}$ was a vector of zeros (model 0 was a threshold model without genetic effects), $\mathbf{a}$ was a vector of additive genetic effects, $\mathbf{d}$ was a vector of dominance genetic effects, $\mathbf{m}$ was a vector of additive genetic mutational effects, and $\mathbf{Z}_2$ was the appropriate matrix of genetic incidences. Under these parameterizations, $\mathbf{a}$ and $\mathbf{d}$ stored the additive and dominance breeding values inherited from the genetic variability in the base generation (Henderson, 1973), whereas $\mathbf{m}$ contained the breeding values originated by additive mutations. Systematic and permanent environmental effects were defined according to Casellas et al. (2007a) based on the same data set. In all models, $\mathbf{b}$ accounted for the age of the ewe (<3, 3 to 5, and >5 yr) and the year of lambing, with 21 levels (from 1986 to 2006; primiparous ewes that lambed in January and February were assigned to the preceding year), and $\mathbf{p}$ was the random permanent environment of the ewe.

Bayesian Inference. The Bayesian development is described for the more complex model (model ADM). The same approach can be assumed for the remaining 7 models after removing the appropriate terms from the Bayesian likelihood and their a priori distributions. Liabilities were assumed to follow a multivariate normal distribution,

$$p(\mathbf{u}|\mathbf{b}, \mathbf{p}, \mathbf{a}, \mathbf{d}, \mathbf{m}) = MVN(\mathbf{Xb} + \mathbf{Z}_1\mathbf{p} + \mathbf{Z}_2\mathbf{a} + \mathbf{Z}_2\mathbf{d} + \mathbf{Z}_2\mathbf{m}, \mathbf{I}),$$

where $\mathbf{I}$ was an identity matrix with dimensions equal to the number of phenotypic records. Note that litter size data were analyzed on a dichotomous scale (single vs. multiple births) and the residual variance was therefore arbitrarily fitted to 1 (Sorensen et al., 1995). Within this context, the contribution to the likelihood of the ith phenotypic observation (y_i) conditional on the ith element in vector $\mathbf{u}$ (u_i) was defined as

$$p(y_i|u_i) = I(y_i = 1) \times I(u_i < 0) + I(y_i = 2) \times I(u_i > 0),$$

where $I(\ )$ was an indicator function with the argument defined in parentheses. The indicator function had a value of 1 if the evaluated expression was true and a value of 0 if not. A priori distributions for **p**, **a**, **d**, and **m** were stated as $p\left(\mathbf{p}\middle|\sigma_p^2\right) = MVN\left(\mathbf{0}, \mathbf{I}\sigma_p^2\right)$, $p\left(\mathbf{a}\middle|\mathbf{A}, \sigma_a^2\right) = MVN\left(\mathbf{0}, \mathbf{A}\sigma_a^2\right)$, $p\left(\mathbf{d}\middle|\mathbf{D}, \sigma_d^2\right) = MVN\left(\mathbf{0}, \mathbf{D}\sigma_d^2\right)$, and $p\left(\mathbf{m}\middle|\mathbf{M}, \sigma_m^2\right) = MVN\left(\mathbf{0}, \mathbf{M}\sigma_m^2\right)$, respectively. Note that **A** is the matrix of additive relationships between individuals (Wright, 1922), **D** is the matrix of dominance relationships between individuals (Hoeschele and VanRaden, 1991; Varona and Misztal, 1999), **M** is the Wray (1990) numerator relationship matrix adapted to accommodate the occurrence of additive mutations in the genome, and σ_p^2, σ_a^2, σ_d^2, and σ_m^2 are the permanent environmental, additive genetic, dominance genetic, and mutational variances, respectively. Three different genetic groups were defined and founders from the same flock of origin were assigned to the same genetic group. Within this context, **A** was constructed through procedures described by Westell et al. (1988). After accounting for the among-founder flock-of-origin genetic differences by the appropriate construction of **A** in the **a** effect, new mutations were free from distortions attributable to founder origins; therefore, **M** was created without considering genetic groups. Flat priors between 0 and $+\infty$ were assumed for variance components, whereas unbounded flat priors were assumed for systematic effects.

Inferences for all the unknown values in the models were made on the relevant marginal posterior distributions by Gibbs sampling (Gelfand and Smith, 1990). Note that the additive genetic effects (**a** and **m**) were parameterized according to the method of Casellas and Medrano (2008) to avoid the need for generic sampling processes (i.e., Metropolis-Hasting sampling) or a large-scale reconstruction of the inverse of the relationship matrix within each sampling iteration. This parameterization assumes the independent assessment of both **a** and **m** effects, whereas the original development proposed by Wray (1990) relied on the joint estimation of **a + m**. More specifically, 3 independent Monte Carlo Markov chains (**MCMC**) were launched for each model, with a length of 500,000 points; the first 10,000 were discarded as burn-in. The burn-in period was evaluated by both visual inspection and the method of Raftery and Lewis (1992).

Comparing Models. Departures in variance components, predicted breeding values, and predicted permanent environmental effects were characterized to highlight differences in model performance. More specifically, correlation coefficients for the predicted genetic merit defined as the joint contribution of all additive genetic effects of the model were estimated, when possible, between all models. Correlation coefficients were also obtained for the **p** effects between all pairs of models. These correlations were calculated between pairs of average posterior estimates. All correlation coefficients were calculated with the CORR procedure (SAS Inst. Inc., Cary, NC). Note that these approaches did not provide evidence in favor of any particular model, but characterized their outputs.

To compare the 8 models objectively, 2 different statistical approaches were used. The deviance information criterion (**DIC**), developed by Spiegelhalter et al. (2002), was calculated for each MCMC. This statistic measures 2 different sources, or types, of model performance in terms of the posterior expectation of the Bayesian deviance (model fit) and the effective number of parameters (model complexity). To account for the threshold structure of the Bayesian likelihood in both models, calculations were performed following the method of Kizilkaya et al. (2003). Models with smaller DIC should be favored by this approach because it indicates a better fit and a lesser degree of model complexity. Differences of at least 3 to 5 DIC units are generally considered statistically relevant (Burnham and Anderson, 1998; Spiegelhalter et al., 2002, 2003).

In addition, the Bayes factor developed by Verdinelli and Wasserman (1995) and Varona et al. (2001), and further adapted to threshold traits by Casellas and Piedrafita (2006), was used to check the statistical relevance of the additive mutational effects by comparing model M with model 0, model AM with model A, model DM with model D, and model ADM with model AD. After appropriate model reparameterization (see the appendix for an example), the Bayes factor between each pair of competing models was calculated by launching 3 additional MCMC with 500,000 elements (with the first 10,000 elements being discarded as burn-in). Note that the **BF** is the ratio of probabilities between 2 alternative models (Kass and Raftery, 1995), where a BF >1 indicates that the numerator model is preferable, whereas a BF <1 suggests that the denominator model is more probable. To categorize the evidence in favor of a given model, the scale of Jeffreys (1984) was adopted, with levels of $1 <$ BF ≤ 3.16 (the favored model not being worth more than a bare mention), $3.16 <$ BF ≤ 10 (substantial evidence), $10 <$ BF ≤ 31.62 (strong evidence), $31.62 <$ BF ≤ 100 (very strong evidence), and BF ≥ 100 (decisive evidence).

RESULTS AND DISCUSSION

Model Comparison

Making comparisons between models is a topic of major interest in animal breeding programs, given the substantial impact that a model can have on predicted breeding values. Within this context, 2 well-known statistics (DIC and correlation coefficients among genetic merit values) were calculated to decide whether the in-

Table 1. Summary of the deviance information criterion (DIC[1]) and Bayes factor for model comparison[2]

	Model β							
Model α	Model 0	Model A	Model D	Model M	Model AD	Model AM	Model DM	Model ADM
Model 0	3,667.30	−17.10	−9.50	−14.80	−16.50	−22.40	−15.40	−24.50
Model A		3,650.20	7.60	2.30	0.60	−5.30	1.70	−7.40
Model D			3,657.80	−5.30	−7.00	−12.90	−5.90	−15.00
Model M	22.7			3,652.50	−1.70	−7.60	−5.90	−9.70
Model AD					3,650.80	−5.90	1.10	−8.00
Model AM		10.9				3,644.90	7.00	−2.10
Model DM			8.5				3,651.90	−9.10
Model ADM					12.5			3,642.80

[1]From Spiegelhalter et al. (2002).

[2]The average DIC for each model is in the diagonal element, whereas DIC differences between model β and model α are in the section above the diagonal. The values in the section below the diagonal are the average Bayes factor of model α (numerator) against model β (denominator).

clusion of additive mutational effects was unnecessary, optional, or essential for an accurate analysis and genetic evaluation of litter size in Ripollesa ewes. Taking average DIC values, this statistic favored model ADM, with a difference of at least 7.40 DIC units with all the remaining models except model AM (−2.10 DIC units). In a similar way, model AM was preferred when compared with models 0, A, D, M, AD, and DM, with a reduction of more than 5 DIC units (Table 1). Although model AM and model ADM did not differ (DIC differences were less than 3 units), they significantly improved model performance in terms of model complexity and fit when compared with the remaining ones (Spiegelhalter et al., 2002, 2003). Although the inclusion of additive genetic terms (**a** and **m**) in the model implied a relevant increase in model parameterization, DIC suggested a sufficient, and even larger, compensation in terms of model fit to litter size data. On the other hand, the relevance of the dominance genetic effects was not completely corroborated by the differences in DIC, although the slight advantage of model ADM suggested that this dominance source of variation could strongly influence litter size data in the Ripollesa sheep breed. When the focus was on the additive genetic mutational effects, the Bayes factors corroborated its statistical relevance in this data set, with models with **m** effects being between 22.7 (model M vs. model 0) and 8.5 times (model DM vs. model D) more plausible than the models excluding this term. These values revealed either substantial or strong evidence for additive genetic mutational effects according to the scale of Jeffreys (1984). Both statistics supported the main idea that mutational effects must be considered in genetic models for litter size evaluation in Ripollesa sheep. Note that models AM and ADM accounted for breeding values in a more flexible way, with the breeding value of a given individual being not only one-half the breeding value of its parents plus a Mendelian sampling contribution, but also including an extra term sampled with the sole restriction of σ_m^2. These models (AM and ADM) could take advantage of this more flexible characterization of breeding values and provide moderate rearrangements in the genetic ranking, with a correlation coefficient of 0.930, mainly in nonextreme individuals (Figure 1). Alternatively, the superiority of mutational models could be partially due to an increase in the genetic variance over time as a scale effect after directional selection for litter size. This hypothesis agreed with the significant and positive genetic trend for litter size reported in this flock (Casellas et al., 2007b). Nevertheless, increasing the inbreeding inherent to closed flocks without foreign contributions must attenuate or even completely remove any increase in genetic variance attributable to scale effects. As noted by Casellas and Medrano (2008), a substantial loss of genetic variance attributable to inbreeding can be anticipated in small populations, competing against the contribution of new mutations.

Predicted Genetic and Permanent Environmental Effects

Results revealed a high correlation for additive genetic effects (from 0.842 to 0.932; $P < 0.001$), although suggesting some rearrangements of individuals according their estimated genetic merit (Figure 1). Indeed, the correlation coefficient varied with respect to parametric space, with the correlation coefficient being significantly greater ($P < 0.05$) for extreme individuals than for the others (Figure 1). Within the context of a breeding scheme, these differences in model performance would not be relevant for rams, given that increased selection pressure was applied and only a few top individuals were selected each year (1 or 2 new rams per year in this flock), whereas they could have had a substantial impact on the performance of selection decisions for ewes, for which between one-quarter and one-third of the female lambs born were kept as replacement individuals. Whatever the case, these discrepancies highlight the need for accurate and objective model comparison when choosing the best model.

Predicted permanent environmental values showed a high and positive correlation between all models (0.902 to 0.985; $P < 0.001$), but without any significant chang-

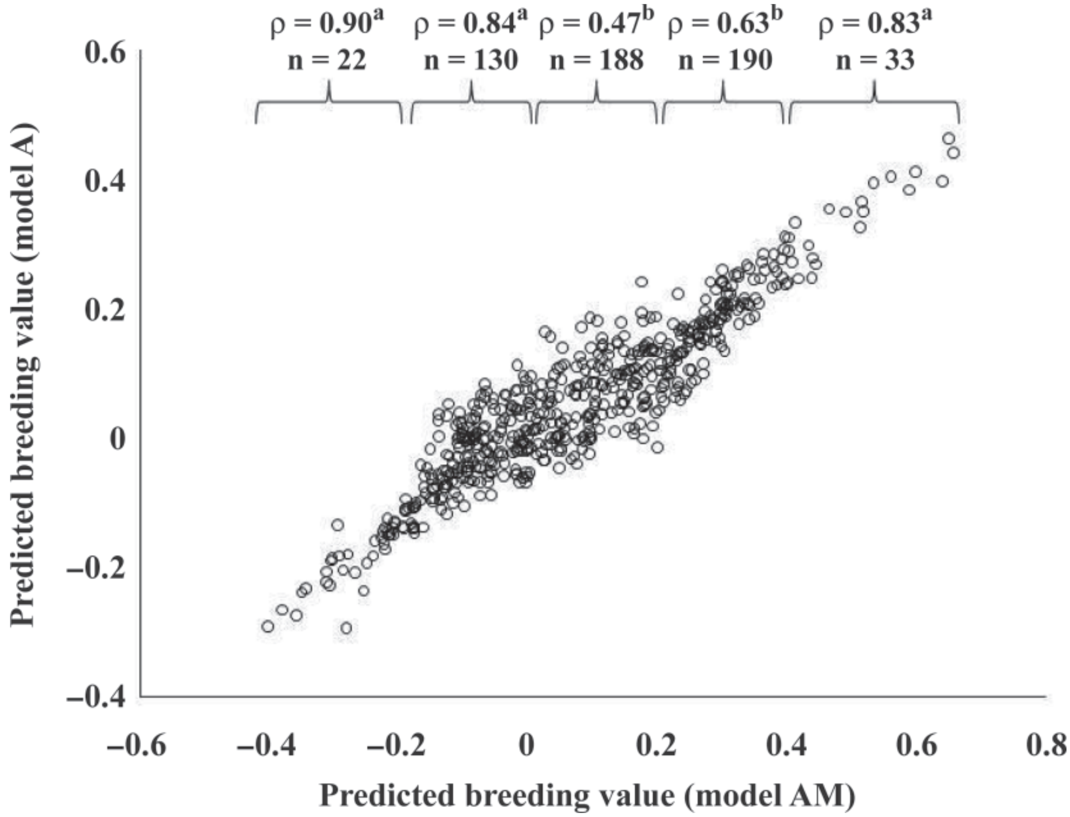

Figure 1. Plot of average estimates for predicted breeding values under model A (standard infinitesimal genetic effect) and model AM (standard infinitesimal genetic effect plus additive genetic mutational effect), and correlation coefficients (ρ) by intervals. Note that correlation coefficients without a common superscript differ significantly ($P > 0.05$).

es within the range of parametric space. The differences in terms of permanent environmental effects between models could therefore have been due to the relocation of part of the unaccounted for genetic effects into the vector **p**. Even so, the permanent environmental effect cannot be considered independently from genetic effects, even in model ADM. Part of the variability that accounted for **p** could have originated from mutations with major additive effects (Casellas and Medrano, 2008), dominance mutations (Zhang et al., 2004), maternal effects (Wu et al., 2006), or epistasis (Noguera et al., 2009).

Variance Components

From this point on, variance components are summarized and discussed on the basis of the 2 models with the smallest DIC, model AM and model ADM. These models revealed slight departures for σ_a^2 and σ_p^2 in terms of modal estimates, although the respective highest posterior density regions at 95% (**HPD95**) overlapped (Table 2) and statistically relevant departures could be discarded. In both cases, the percentage of phenotypic variance accounted for by σ_a^2 was approximately 13%, matching previous estimates obtained with the same data set (Casellas et al., 2007b,c). These values were within the range of estimates provided by Fogarty (1995) and were close to the values obtained for the Rasa Aragonesa (8%; Altarriba et al., 1998) and Finnsheep (14%; Matos et al., 1997) breeds. The percentages of phenotypic variance accounted for by σ_p^2 were 7.2% (model AM) and 6.8% (model ADM; Table 2), which are smaller estimates than the values reported by Casellas et al. (2007b,c). These departures from the previous estimates on the same data set must be related to the inclusion of additional genetic terms in the model. The additive mutational and the genetic dominance effects could partially account for a part of the effects accumulated in the permanent environmental vector in the study by Casellas et al. (2007c). The dominance variance was small and its HPD95 did not discard the null estimate. This result agreed with the nonsignificant differences between model AM and model ADM in terms of DIC, although it suggested the possibility of some dominance genetic contributions to the litter size data. The dominance variance accounted for 3.5% of the phenotypic variance under model ADM and agreed with the estimate obtained by Culbertson et al. (1998) in Yorkshire swine, although we lack comparable results in sheep.

Both model AM and model ADM revealed relevant uploading of new additive genetic variability originating from mutation. The model estimates for σ_m^2 were 0.013 and 0.008, respectively, with their HPD95 dis-

Table 2. Variance components for litter size in the Ripollesa ewe under model ADM (threshold animal model with additive genetic, dominance genetic, and additive mutational effects)

	Model ADM[2]		Model AM[3]	
Parameter[1]	Mode	HPD95[4]	Mode	HPD95
σ^2_p	0.092	0.008 to 0.243	0.088	0.008 to 0.241
σ^2_a	0.168	0.026 to 0.316	0.162	0.024 to 0.303
σ^2_d	—[5]	—	0.046	0.000 to 0.425
σ^2_m	0.013	0.007 to 0.044	0.008	0.004 to 0.038
c^2	0.072	0.029 to 0.151	0.068	0.029 to 0.162
h^2_a	0.131	0.025 to 0.238	0.124	0.012 to 0.208
h^2_d	—	—	0.035	0.000 to 0.132
h^2_m	0.009	0.005 to 0.034	0.006	0.003 to 0.029

[1] σ^2_p = permanent environmental variance; σ^2_a = additive genetic variance; σ^2_d = dominance genetic variance; σ^2_m = mutational variance; $c^2 = \sigma^2_p / \sigma^2_T$; $h^2_a = \sigma^2_a / \sigma^2_T$; $h^2_d = \sigma^2_d / \sigma^2_T$; $h^2_m = \sigma^2_m / \sigma^2_T$; $\sigma^2_T = \sigma^2_p + \sigma^2_a + \sigma^2_d + \sigma^2_m + 1$.

[2] Threshold animal model accounting for additive genetic effects, dominance genetic effects, and additive genetic mutational effects.

[3] Threshold animal model accounting for additive genetic effects and additive genetic mutational effects.

[4] HPD95 = highest posterior density region at 95%.

[5] A dash indicates not estimable.

carding the null estimate (Table 2). Within this context, mutational heritability values were 9.0×10^{-3} and 6.0×10^{-3}, respectively, and fell within the range of mutational heritability values summarized by Lynch (1988) and Houle et al. (1996) in experimental species (10^{-2} to 5×10^{-4}). These values were similar to those obtained for reproductive traits in other species, such as for litter size in mice (8.0×10^{-3}; Casellas and Medrano, 2008), fecundity in *Drosophila melanogaster* (3.4×10^{-3}; Houle et al., 1994), and clutch size in *Daphnia pulex* (1.4×10^{-3}; Lynch, 1985), and were in line with previous research findings that reported direct evidence of new mutations with major effects in sheep, such as the Booroola (Wilson et al., 2001) and callipyge phenotypes (Davis et al., 2004). Nevertheless, the estimated mutational heritability values for litter size in the Ripollesa sheep were close to the greater boundary of the range summarized by Lynch (1988) and Houle et al. (1996), suggesting that these σ^2_m estimates could be partially influenced by other sources of variation. Model ADM showed a moderate reduction in the model estimate of σ^2_m when compared with model AM, suggesting that some dominance effects were accounted for by the additive mutational effect in model AM. In a similar way, other genetic influences, such as maternal effects from the dam of the ewe or the generation of new mutations with large effects, could be suggested as potential distorters. Nevertheless, the analytical models included a permanent environmental effect to account for these additional unpredictable sources of variation, and the estimated variance components must be free from significant biases.

To our knowledge, this is the first estimate of h^2_m in a livestock species. This suggests that mutation-related theoretical developments (Hill, 1982a,b) could also be extrapolated to farm animals. The appearance of new mutations is a phenomenon of particular interest in local and endangered sheep populations, such as the Ripollesa breed. The substantial loss of genetic diversity attributable to inherent inbreeding in small flocks, together with limited contact between different flocks and the small number of rams contributing to each new generation, (Norberg and Sørensen, 2007; Casellas et al., 2009) could be partially balanced by mutation. Nevertheless, further studies are required to characterize the balance between inbreeding and mutation in Ripollesa sheep.

LITERATURE CITED

Altarriba, J., L. Varona, L. A. García-Cortés, and C. Moreno. 1998. Bayesian inference of variance components for litter size in Rasa Aragonesa sheep. J. Anim. Sci. 76:23–28.

Burnham, K. P., and D. R. Anderson. 1998. Model Selection and Inference. A Practical Information Theoretic Approach. Springer-Verlag, New York, NY.

Caballero, A., M. A. Toro, and C. Lopez-Fanjul. 1991. The response to artificial selection from new mutations in *Drosophila melanogaster.* Genetics 127:89–102.

Casellas, J., G. Caja, A. Ferret, and J. Piedrafita. 2007a. Analysis of litter size and days to lambing in the Ripollesa ewe. I. Comparison of models with linear and threshold approaches. J. Anim. Sci. 85:618–624.

Casellas, J., G. Caja, A. Ferret, and J. Piedrafita. 2007b. Analysis of litter size and days to lambing in the Ripollesa ewe. II. Estimation of variance components and response to phenotypic selection on litter size. J. Anim. Sci. 85:625–631.

Casellas, J., G. Caja, and J. Piedrafita. 2007c. Genetic determinism for within-litter birth weight variation and its relationship with litter weight and litter size in the Ripollesa ewe breed. Animal 1:637–644.

Casellas, J., G. Caja, X. Such, and J. Piedrafita. 2007d. Survival analysis from birth to slaughter of Ripollesa lambs under semi-intensive management. J. Anim. Sci. 85:512–517.

Casellas, J., and J. F. Medrano. 2008. Within-generation mutation variance for litter size in inbreed mice. Genetics 179:2147–2155.

Casellas, J., and J. Piedrafita. 2006. Bayes factor for testing the genetic background of quantitative threshold traits. J. Anim. Breed. Genet. 123:301–306.

Casellas, J., J. Piedrafita, G. Caja, and L. Varona. 2009. Analysis of founder-specific inbreeding depression on birth weight in Ripollesa lambs. J. Anim. Sci. 87:72–79.

Culbertson, M. S., J. W. Mabry, I. Misztal, N. Gengler, J. K. Bertrand, and L. Varona. 1998. Estimation of dominance variance in purebred Yorkshire swine. J. Anim. Sci. 76:448–451.

Davis, E., C. H. Jensen, H. D. Schroder, F. Farnir, T. Shay-Hadfield, A. Kliem, N. Crockett, M. Georges, and C. Charlier. 2004. Ectopic expression of DKL1 protein in skeletal muscle of padumnal heterozygotes causes the callipyge phenotype. Curr. Biol. 14:1858–1862.

Fogarty, N. M. 1995. Genetic parameters for live weight, fat and muscle measurements, wool production and reproduction in sheep: A review. Anim. Breed. Abstr. 63:101–143.

García-Cortés, L. A., C. Cabrillo, C. Moreno, and L. Varona. 2001. Hypothesis testing for the genetic background of quantitative traits. Genet. Sel. Evol. 33:3–16.

Gelfand, A., and A. F. M. Smith. 1990. Sampling based approaches to calculating marginal densities. J. Am. Stat. Assoc. 85:398–409.

Goddard, M. E. 2001. The validity of genetic models underlying quantitative traits. Livest. Prod. Sci. 72:117–127.

Henderson, C. R. 1973. Sire evaluation and genetic trends. Pages 10–41 in Proc. Anim. Breeding Genet. Symp. in Honor of Dr. Jay L. Lush. Am. Soc. Anim. Sci., Champaign, IL.

Hill, W. G. 1982a. Prediction of response to artificial selection from new mutations. Genet. Res. 40:255–278.

Hill, W. G. 1982b. Rates of change in quantitative traits from fixation of new mutations. Proc. Natl. Acad. Sci. USA 79:142–145.

Hill, W. G. 2007. A century of corn selection. Science 307:683–684.

Hoeschele, I., and P. M. VanRaden. 1991. Rapid inversion of dominance relationship matrices for noninbred populations by including sire by dam subclass effects. J. Dairy Sci. 74:557–569.

Houle, D., K. A. Huges, D. K. Hoffmaster, J. Ihara, S. Assimacopoulos, D. Canada, and B. Charlesworth. 1994. The effects of spontaneous mutation on quantitative traits. I. Variances and covariances of life history traits. Genetics 138:773–785.

Houle, D., B. Morikawa, and M. Lynch. 1996. Comparing mutational variabilities. Genetics 143:1467–1483.

Jeffreys, H. 1984. Theory of Probability. 3rd ed. Clarendon Press, Oxford, UK.

Kambadur, R., M. Sharma, T. P. L. Smith, and J. J. Bass. 1997. Mutations in *myostatin* (*GDF8*) in double-muscled Belgian Blue and Piedmontese cattle. Genome Res. 7:910–915.

Kass, R. E., and A. E. Raftery. 1995. Bayes factors. J. Am. Stat. Assoc. 90:773–795.

Keightley, P. D. 1998. Genetic basis of response to 50 generations of selection on body weight in inbred mice. Genetics 148:1931–1939.

Keightley, P. D., and W. G. Hill. 1992. Quantitative genetic variation in body size of mice from new mutations. Genetics 131:693–700.

Kizilkaya, K., P. Carnier, A. Albera, G. Bittante, and R. J. Tempelman. 2003. Cumulative t-link threshold models for the genetic analysis of calving ease scores. Genet. Sel. Evol. 35:489–512.

Lynch, M. 1985. Spontaneous mutations for life-history characters in an obligate parthenogen. Evolution 39:804–818.

Lynch, M. 1988. The rate of polygenic mutation. Genet. Res. 51:137–148.

MacLennan, D. H., and M. S. Phillips. 1992. Malignan hyperthermia. Science 256:789–794.

Matos, C. A. P., D. L. Thomas, D. Gianola, R. J. Tempelman, and L. D. Young. 1997. Genetic analysis of discrete reproductive traits in sheep using linear and nonlinear models: I. Estimation of genetic parameters. J. Anim. Sci. 75:76–87.

Noguera, J. L., M. C. Rodríguez, L. Varona, A. Tomàs, G. Muñoz, O. Ramírez, C. Barragán, M. Arqué, J. P. Bidanel, M. Amills, and A. Sánchez. 2009. A bi-dimensional genome scan for prolificacy traits in pigs shows the existence of multiple epistatic QTL. BMC Genomics 10:636. doi:10.1186/1471-2164-10-636

Norberg, E., and A. C. Sørensen. 2007. Inbreeding trend and inbreeding depression in the Danish populations of Texel, Shropshire, and Oxford Down. J. Anim. Sci. 85:299–304.

Olesen, I., M. Pérez-Enciso, D. Gianola, and D. L. Thomas. 1994. A comparison of normal and nonnormal mixed models for number of lambs born in Norwegian sheep. J. Anim. Sci. 72:1166–1173.

Raftery, A. E., and S. M. Lewis. 1992. How many iterations in the Gibbs sampler? Pages 763–773 in Bayesian Statistics IV. J. M. Bernardo, J. O. Berger, A. P. Dawid, and A. F. M. Smith, ed. Oxford Univ. Press, Oxford, UK.

Sorensen, D. A., S. Andersen, D. Gianola, and I. Korsgaard. 1995. Bayesian inference in threshold models using Gibbs sampling. Genet. Sel. Evol. 27:229–249.

Spiegelhalter, D. J., N. G. Best, B. P. Carlin, and A. van der Linde. 2002. Bayesian measures of model complexity and fit (with discussion). J. R. Stat. Soc., B 64:583–639.

Spiegelhalter, D. J., A. Thomas, N. G. Best, and D. Lunn. 2003. WinBUGS Version 1.4 User Manual. MRC Biostatistics Unit, Cambridge, UK. http://www.mrc-bsu.cam.ac.uk/bugs/ Accessed Feb. 6, 2010.

Van Tassell, C. P., L. D. Van Vleck, and K. E. Gregory. 1998. Bayesian analysis of twining and ovulation rates using a multiple-trait threshold model and Gibbs sampling. J. Anim. Sci. 76:2048–2061.

Varona, L., L. A. García-Cortés, and M. Pérez-Enciso. 2001. Bayes factors for detection of quantitative trait loci. Genet. Sel. Evol. 33:133–152.

Varona, L., and I. Misztal. 1999. Prediction of parental dominance combinations for planned matings, methodology, and simulation results. J. Dairy Sci. 82:2186–2191.

Verdinelli, I., and L. Wasserman. 1995. Computing Bayes factors using a generalization of the Savage-Dickey density ratio. J. Am. Stat. Assoc. 90:614–618.

Waldron, D. F., and D. L. Thomas. 1992. Increased litter size in Rambouillet sheep. I. Estimation of genetic parameters. J. Anim. Sci. 70:3333–3344.

Wang, C. S., J. J. Rutledge, and D. Gianola. 1994. Bayesian analysis of mixed linear models via Gibbs sampling with an application to litter size in Iberian pigs. Genet. Sel. Evol. 26:91–115.

Westell, R. A., R. L. Quaas, and L. D. Van Vleck. 1988. Genetic groups in an animal model. J. Dairy Sci. 71:1310–1318.

Wilson, T., X.-Y. Wu, J. L. Juengel, I. K. Ross, J. M. Lumsden, E. A. Lord, K. G. Dodds, G. A. Walling, J. C. McEwan, A. R. O'Connell, K. P. McNatty, and G. W. Montgomery. 2001.

Highly prolific Booroola sheep have a mutation in the intracellular kinase domain of bone morphogenetic protein 1B Receptor (ALK-6) that is expressed in both oocytes and granulose cells. Biol. Reprod. 64:1225–1235.
Wray, N. R. 1990. Accounting for mutation effects in the additive genetic variance-covariance matrix and its inverse. Biometrics 46:177–186.
Wright, S. 1922. Coefficients of inbreeding and relationship. Am. Nat. 56:330–338.
Wright, S. 1934. An analysis of variability in number of digits in an inbred strain of Guinea pigs. Genetics 19:506–536.
Wu, G., F. W. Bazer, J. M. Wallace, and T. E. Spencer. 2006. Board-Invited Review: Intrauterine growth retardation: Implications for the animal sciences. J. Anim. Sci. 84:2316–2337.
Zhang, X. S., J. Wang, and W. G. Hill. 2004. Influence of dominance, leptokurtosis and pleiotropy of deleterious mutations on quantitative genetic variation at mutation-selection balance. Genetics 166:597–610.

APPENDIX

Model ADM Reparameterization for Bayes Factor Calculation

Following the methods of García-Cortés et al. (2001) and Varona et al. (2001), the Bayes factor between model ADM and model AD can be computed by analyzing the more complex model (model ADM) after appropriate reparameterization. Models AD and ADM for litter size analyses in Ripollesa sheep had been previously typified as $\mathbf{u} = \mathbf{Xb} + \mathbf{Z}_1\mathbf{p} + \mathbf{Z}_2\mathbf{a} + \mathbf{Z}_2\mathbf{d} + \mathbf{e}$ and $\mathbf{u} = \mathbf{Xb} + \mathbf{Z}_1\mathbf{p} + \mathbf{Z}_2\mathbf{a} + \mathbf{Z}_2\mathbf{d} + \mathbf{Z}_2\mathbf{m} + \mathbf{e}$, respectively. As described above, the mutation effects and residuals were assumed to be normally distributed:

$$p\left(\mathbf{m}\middle|\mathbf{M},\sigma_m^2\right) = MVN\left(\mathbf{0},\mathbf{M}\sigma_m^2\right), \text{ and}$$

$$p(\mathbf{e}) = MVN(\mathbf{0}, \mathbf{1}),$$

where the residual variance was arbitrarily fitted to 1 (Sorensen et al., 1995). These assumptions allowed a straightforward reparameterization of model ADM (model ADM*) as

$$\mathbf{u} = \mathbf{Xb} + \mathbf{Z}_1\mathbf{p} + \mathbf{Z}_2\mathbf{a} + \mathbf{Z}_2\mathbf{d} + \mathbf{e^*}.$$

The new residual term ($\mathbf{e^*}$) absorbed the mutational genetic effects and distributed them as follows:

$$p(\mathbf{e^*}|\mathbf{V}) = MVN(\mathbf{0}, \mathbf{V}),$$

where $\mathbf{V} = \sigma_{e*}^2\left[\mathbf{Z}_2\mathbf{M}\mathbf{Z}_2'\rho^2 + \mathbf{I}\left(1-\rho^2\right)\right]$ was the residual (co)variance matrix, $\sigma_{e*}^2 = \sigma_m^2 + 1$ was the variance of $\mathbf{e^*}$, and $\rho^2 = \sigma_m^2 / \sigma_{e*}^2$ was the intraclass correlation. At this point, a priori distributions for liabilities and ρ^2 had to be rewritten appropriately as

$$p\left(\mathbf{b},\mathbf{p},\mathbf{a},\mathbf{d},\sigma_{e*}^2,\rho^2\middle|\mathbf{u}\right) = MVN\left(\mathbf{Xb} + \mathbf{Z}_1\mathbf{p} + \mathbf{Z}_2\mathbf{a} + \mathbf{Z}_2\mathbf{d},\mathbf{V}\right), \text{ and}$$

$$p\left(\rho^2\right) = k \quad \text{if } \rho^2 \in \left[0,\ 1\right] \text{ and 0 otherwise.}$$

A flat prior was assumed for σ_{e*}^2, whereas the remaining $p(\mathbf{y}|\mathbf{u})$, $p\left(\mathbf{p}\middle|\sigma_p^2\right)$, $p\left(\mathbf{a}\middle|\sigma_a^2\right)$, $p\left(\mathbf{d}\middle|\sigma_d^2\right)$, $p\left(\sigma_p^2\right)$, $p\left(\sigma_a^2\right)$, and $p\left(\sigma_d^2\right)$ densities were expressed as in the standard model ADM (see above). Note that model ADM* and model AD differed in only 1 bounded variable (ρ^2) and that the Bayes factor was reduced to

$$\mathrm{FB}_{\mathrm{ADM/AD}} = \frac{p\left(\rho^2 = 0\right)}{p\left(\rho^2 = 0\middle|\mathbf{y}\right)} = \frac{1}{p\left(\rho^2 = 0\middle|\mathbf{y}\right)},$$

where $p(\rho^2 = 0|\mathbf{y})$ sufficed to obtain the ratio of probabilities between the 2 competing models. As described by Varona et al. (2001), $p(\rho^2 = 0|\mathbf{y})$ can be straightforwardly obtained from the Gibbs sampler output of model ADM* by averaging the full conditional densities of each cycle at $\rho^2 = 0$ using the Rao-Blackwell argument (Wang et al., 1994). The conditional data augmentation step described by Casellas and Piedrafita (2006) is required to update liabilities appropriately within each Gibbs sampling cycle.